FRA Group Project 1

This project puts to real use concepts like VaR, Markowitz Frontier, Expected Shortfall etc.

Every portfolio manager makes a risk call "this one's safe, this one's aggressive" based on today's data. But would that call have held up in a crash? And would today's "optimal" mix still look optimal if you rebuilt it using a different era's numbers? That's what you're going to find out, with your own ₹15 lakh of imaginary money and your own stocks.

What you'll build

Two portfolios of at least 10 stocks each, no stock repeated across both:

- **Portfolio A – High Risk** (going for return)
- **Portfolio B – Low Risk** (playing it safe)

₹15L into each, and yes, that number should be easy to change if you want to experiment.

What you'll do

1. Make the call. Decide why A is high risk and B is low risk. Use whatever lens makes sense to you (e.g. volatility, beta, market cap, sector, ratings) but just back it with a number and logic, not just a vibe.

2. Time-travel test #1: The risk label. Pick two moments in history: one a crisis, one calm. Your choice, your reasoning. Run VaR and Expected Shortfall for both portfolios in both periods and see: did "high risk" actually behave like high risk back then? Did "low risk" hold up? You are NOT restricted to only VaR and ES. Feel free to use additional measures but justify.

3. Time-travel test #2: The allocation of stocks in the portfolio. Take the return your portfolio earns today, and ask: if you rebuilt the efficient frontier using each historical period's data, aiming for that same return, would the optimizer still pick your weights? Or would it build something different?

Put it on display

Build a dashboard (Use your own creativity on structure, features etc) hosted, link-shareable, something you'd genuinely put on a resume. Make it something a recruiter could open and understand quickly

Presentations

15 minutes maximum presentation time